Exp no: 02 Study of different types of Network Cables

Date: 27/7/26

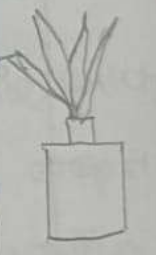
Aim:

Understand different types of network cables

different type of cables used in networking are

1. Unshield Twisted Pair (UTP) Cable
2. Shield Twisted Pair (STP) Cable
3. Coaxial Cable
4. Fibre Optic cable

Cable Type

Cable Type	Category	Maximum data transmission	Advantage/disadvantage	Application	Image
UTP	Category 5	upto 100 Mbps	Adv:- * Cheaper in cost DisAdv:- * More prone to (EMI) and noise	10 Base - T Ethernet Fast Ethernet, Gigabit Ethernet	
	Category 5e	1 Gbps		Fast Ethernet, gigabit Ethernet	
STP	category 6, 6a	10 Gbps	Adv:- * Shielded * Faster than UTP Dis adv:- * Expensive * greater installation	• Gigabit Ethernet, • 10 G Ethernet (5m) widely used in data centers gigabit Ethernet, 10 G Ethernet	
SSTP	Category 7	10 Gbps			
Coaxial Cable	RG-6 RG-59 RG-11	10-100 Mbps	Adv * High bandwidth * Low loss bandwidth Disadv:- * Limited distance * Cost	Speed of signal is 500m, Television network, High speed internets	
fibre optics Cable	Single mode Multimode	100 Gbps	Adv:- * High Speed * High Bandwidth Disadv:- * Expensive * Requires skilled	Maximum distance of fibre optics cable is around 100 meters	

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Result :

The Types of network cables and making cross wired cable & straight through cables using clamping / crimping tool is studied